

Instructor: Prof. Michael Sprague
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Discussion Section Leader: Mr. John Loffeld
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Lecture Time & Location: MWF, 11:00 am – 11:50 am, COB 113

Discussion Section Time & Location:
001: Thurs. 12:00 pm – 1:50 pm, COB 127
002: Thurs. 2:00 pm – 3:50 pm, COB 127

Office hours:
Wednesday: Sprague, 1:30–2:30 pm (S&E 332)
Thursday: Loffeld, 10:00 am – noon (Library table, fourth floor)
Friday: Sprague, 9:30 – 10:30 am (S&E 332)
Alternatively, e-mail or call for appointment.

Course Goals: By the end of this course, I hope you will learn the fundamentals of ordinary differential equations (ODEs) and linear algebra as it applies to ODEs. An important component of the course deals with modeling phenomena in terms of ODEs. While modeling is covered in detail in other courses, it is essential for understanding ODEs and what they represent in real world phenomena.

Topics covered: Solution of first-order ODE's by analytical, graphical and numerical methods; Linear ODE's, especially second order with constant coefficients; Undetermined coefficients and variation of parameters; Sinusoidal and exponential signals: oscillations, damping, resonance; Linear algebra: Gaussian Elimination, basis and dimension, eigenvalues and eigenvectors, complex exponentials; Nonlinear phenomena: limit cycles and chaos.

Textbook: *Differential Equations and Linear Algebra*, by Farlow, Hall, McDill and West. We will cover most of Chapters 1-7.

Course webpage: The Math 24 website is part of the UCMCROPS course management system (ucmcrops.ucmerced.edu). It is available automatically to all students enrolled in this class. The website contains the course calendar, announcements, and email list. We will use this site for distributing course materials.

Discussion section: Discussion sections meet for two hours on Thursdays, where you will develop and practice your problem solving skills by working with your classmates to solve challenging problems. *Your participation in the discussion sections is graded* (see Grade determination below).

Grade determination: Your final grade in the course will be based on a total of 600 points: homework assignments (75 points), discussion-section participation (75 points), three midterm exams (100 points each), and a cumulative final exam (150 points). If you obtain 90% of the total points, you will definitely receive an A in the course. If you obtain less than 55% of the total points, you will definitely receive an F. For everything in between, letter grades will be given using the **approximate** framework: A: 90-100%, B: 80-90%, C: 70-80%, D: 60-70%. Please be aware that you need a C– or better to proceed to Advanced Mathematical Methods as well as advanced Engineering courses.

Homework: Homework is given after each lecture and is typically due at the **start** of the Friday lecture (see class schedule). Late homework will **not** be accepted or graded. 50% of each homework score will be based on completion and presentation; 50% will be based on the correctness of one or two of the assigned problems. Graded homework will be returned during the next discussion section. Your lowest two homework scores will be dropped when determining your final grade.

You are encouraged to work in groups. However, **all work turned in must be your own**. At the end of your written homework, you must identify explicitly all individuals with whom you worked for each problem. You must also **list explicitly any outside sources employed** (e.g. websites, *Mathematica*, book other than

the textbook, etc.) for each problem you solve. This does not mean that you are allowed to copy a solution should you find it posted elsewhere (see Academic integrity below).

Homework must be presented according to the guidelines at the end of this document. Only the first page of *unstapled* multiple-page homework will be graded.

Exams: There will be three unit (mid-term) exams and a comprehensive final. The lecture on the Wednesday and Discussion Section on Thursday before each exam will be a review session. The dates and times for the exams are

exam 1: Thursday, Sept. 28 6:00–7:30 pm

exam 2: Thursday, Oct. 26 6:00–7:30 pm

exam 3: Thursday, Nov. 30 6:00–7:30 pm

final exam: Tuesday, Dec. 12 8:00–11:00 am

Math 24 exams from previous semesters may be found at <http://appliedmath.ucmerced.edu/exams.html>

There will be no make-up exams or early exams! If you are sick during a unit exam, please bring a note from your doctor verifying your illness. Your course grade will then be determined by the rest of your course work. Please bring your student ID to each exam. Calculators are not allowed on the exams. A special needs room for people with documented disabilities will be provided for each exam. Speak to me for more information.

Crib sheet: One double-sided letter-size crib sheet is allowed. Note: your crib sheet must remain in your possession and cannot be shared with others.

Cell phones: All portable electronic devices (e.g., cell phones & pagers) must be turned off and put away during exams, lectures, and discussion sections. Calculators are the exception; they may be used in lectures and discussion sections, but **not** in exams.

Calculators and computers: You may use a calculator (graphing or otherwise) or other computational tool (e.g., Mathematica, Maple, Matlab, etc) to aid in your completion of homework assignments. A free alternative, which is well suited for this course, is the Mathematical Visualization Toolkit (MVT)¹. You will need some calculation tools for certain homework and discussion section problems. Remember, however, that there will be no calculators allowed in the exams.

Dropping the course: You may drop this course without paying a fee and without further approval before 5:00 pm, Friday, Sept. 22. Dropping the course after this time, but before 5:00 pm, Friday, Nov. 3, requires the signed approval of the instructor, and the confirmation of the Dean of the School of Natural Sciences. Students may not drop after Friday, Nov. 3 at 5:00 pm. Please see the UC Merced *General Catalog* for more details.

Extra help: You are encouraged to get extra help whenever you need it. Instructor and TA office hours are posted on the course web page. Other helpful items are posted on the UCMCROPS page. You are welcome to send questions to me via e-mail at any time.

Blue books: Each student is required to purchase **four** blue books and give them to the TA by the beginning of the Discussion Section on Thursday, September 7. These will be distributed for the exams, so please do not write anything (not even your name) on the blue books.

Special accommodations: If you qualify for accommodations because of a disability, please submit a letter from Disability Services to the instructor in a timely manner so that your needs may be addressed. Student Affairs determines accommodations based on documented disabilities.

I will make every effort to accommodate all students who, because of religious obligations, have conflicts with scheduled exams, assignments, or required attendance. Please speak with me during the first week of classes regarding any potential academic adjustments or accommodations that may arise due to religious beliefs during this term.

¹available at <http://amath.colorado.edu/java/index.php>

Academic integrity: Academic integrity is the foundation of an academic community and without it none of the educational or research goals of the university can be achieved. All members of the university community are responsible for its academic integrity. Existing policies forbid cheating on examinations, plagiarism and other forms of academic dishonesty. The current policies for UC Merced are described in the Academic Honesty Policy (see under Student Judicial Affairs at <http://studentlife.ucmerced.edu>) The following general guidelines are adapted from the UC Davis Code of Academic Conduct (<http://sja.ucdavis.edu/cac.html>).

Examples of academic dishonesty include:

- receiving or providing unauthorized assistance on examinations
- using unauthorized materials during an examination
- plagiarism - using materials from sources without citations
- altering an exam and submitting it for re-grading
- fabricating data or references
- using false excuses to obtain extensions of time or to skip coursework

The ultimate success of a code of academic conduct depends largely on the degree to which the students fulfill their responsibilities towards academic integrity. These responsibilities include:

- Be honest at all times.
- Act fairly toward others. For example, do not disrupt or seek an unfair advantage over others by cheating, or by talking or allowing eyes to wander during exams.
- Take group as well as individual responsibility for honorable behavior. Collectively, as well as individually, make every effort to prevent and avoid academic misconduct, and report acts of misconduct which you witness.
- Do not submit the same work in more than one class. Unless otherwise specified by the instructor, all work submitted to fulfill course requirements must be work done by the student specifically for that course. This means that work submitted for one course cannot be used to satisfy requirements of another course unless the student obtains permission from the instructor.
- Unless permitted by the instructor, do not work with others on graded coursework, including in class and take-home tests, papers, or homework assignments. When an instructor specifically informs students that they may collaborate on work required for a course, the extent of the collaboration must not exceed the limits set by the instructor.
- Know what plagiarism is and take steps to avoid it. When using the words or ideas of another, even if paraphrased in your own words, you must cite your source. Students who are confused about whether a particular act constitutes plagiarism should consult the instructor who gave the assignment.
- Know the rules – ignorance is no defense. Those who violate campus rules regarding academic misconduct are subject to disciplinary sanctions, including suspension and dismissal.

Homework preparation requirements & guidelines

(taken from <http://www.math.hmc.edu/teaching/homework/>):

Learning mathematics involves learning how to communicate your ideas effectively. As a student, much of this communication will be in the form of homework. Therefore, so that we may provide you with meaningful and worthwhile feedback, it is important that you put your homework in an easy to read, easy to navigate format. After all, how you present your work should enhance the ideas you are trying to communicate, not impede them.

It is good practice to first work out the solutions to homework problems on scratch paper, and to then neatly write up your solutions. This will help you to turn in a clean finished product.

The following are the **requirements** for submitting homework in Math 24:

- Your handwriting should be legible.
- Homework with multiple pages must be stapled in the upper left-hand corner.
- In the upper right-hand corner you must write (in this order)
 - First and Last Name
 - “Math 24”, followed by your Discussion Section Number
 - Homework Set Number
 - Due Date of the Homework
- Problems should be clearly labeled and numbered on the left-hand side of the page. There should also be a visible separation between problems.
- Problems should be written in the order they are assigned.
- Any graphs should have clearly labeled axes
- At the end of your written homework, you must explicitly identify all individuals with whom you worked. You must also **explicitly list any outside sources employed** (websites, *Mathematica*, book other than course text, etc.).